

How Often Should We Re-Optimize?

The next question that often arises is how often should we re-optimize the portfolio? Many of the self-proclaimed industry pundits state it should be done continuously. Others state that re-optimization should be performed at certain intervals, such as every 5, 10, or 15 minutes. We could not disagree more. Portfolio algorithms should be re-optimized if performance is projected to be dramatically different than what is expected, market conditions are different than what we planned, or if there are opportunities to take advantage of liquidity and prices. The difference between single stock and portfolio algorithms, however, is that when trading portfolios investors are interested in the overall portfolio risk and not necessarily the performance of an individual order. For example, if there is sufficient liquidity and favorable prices that would allow the trader to complete the order at great prices it may not be in their best interest to do so if the net result would adversely affect the hedge of the portfolio and increase overall portfolio risk. Portfolio analysis needs to be performed from the risk perspective of the portfolio not from the risk perspective of any individual stock.

Our recommendation for re-optimization criteria is based on the Z-score of projected performance. This is similar to the Z-score re-optimization criteria used for the single stock algorithms. This measure is:

$$Z_t = \frac{E_0(\text{Cost}) - E_t(\text{Cost})}{E_t(\text{Timing Risk})} \quad (9.24)$$

A positive score indicates investors are performing better than projected and a negative score indicates investors are performing worse than projected. The Z-score above measures the number of standard deviations away from our original cost estimate we are projected to finish given actual market conditions. Investors could elect to re-optimize the portfolio algorithm if the Z-score at any point in time exceeds a specified range such as $|Z| > \pm 1$ or $|Z| > \pm \frac{1}{2}$. Some investors elect to only re-optimize if the Z-score is less than a specified value. In these cases re-optimization would only occur if performance is expected to be less favorable.

Investors, of course, could also re-optimize if there is opportunity to deviate from an optimally prescribed strategy to improve the overall risk characteristics of the trade list and reduce trading costs. This can also be done on an individual stock basis and is described below.

Investors should also re-optimize and change their strategy if there is reason to believe that their trading intentions have been uncovered by market participants, which would lead to higher trading costs.